Supplementary Material
A Computable Map of Treatment-Sequencing Evidence in HR+/HER2- Metastatic Breast Cancer

H.-T. Lin

May 16, 2026

Supplementary Table S1. Reporting Transparency Checklist

This study reports a graph-theoretic computational analysis of clinical trials and guidelines. No standard reporting guideline (TRIPOD, STROBE, PRISMA, CONSORT, STARD) is a perfect fit. We adopt a custom transparency checklist that combines PRISMA-2020 reporting items (relevant to evidence synthesis) with a graph-encoding addendum specific to the present method.

Item	Reported in	Notes
1. Title identifies study as evidence-synthesis	Title	"treatment-sequencing evidence"
2. Structured abstract	Abstract	300-word JCO CCI structure
3. Rationale	Introduction	Composition across non-overlapping trials
4. Objectives	Introduction	EFDPR + ODI
5. Eligibility criteria for trials	Methods 2.1, 2.2	16-trial pilot drawn from ESMO/ASCO reference lists
6. Information sources	Methods 2.1	ClinicalTrials.gov v2 + ESMO 2024
7. Search strategy	Methods 2.1	Per-NCT fetch list (Supplementary Table S2)
8. Selection process	Methods 2.2	Hand-curated against canonical primary publications
9. Data extraction process	Methods 2.2	Schema v1.0 (frozen)
10. Data items	Methods 2.2	Schema fields per Table S2
11. Risk-of-bias assessment	Discussion	Reviewer-cycle-disclosed corpus-selection bias
12. Effect measures	Methods 2.5	EFDPR; ODI
13. Synthesis methods	Methods 2.5	Concordance algorithm
14. Reporting bias assessment	Limitations	Pilot-scale only
15. Certainty assessment	Discussion	Pre-registered tolerance levels
16. Study selection results	Results 3.1	16/16 trials retrieved
17. Study characteristics	Table S2	Hand-curated schema fills
18. Risk of bias	Limitations	Disclosed
19. Results of individual studies	DAG edges	Trial primary endpoint per edge
20. Results of syntheses	Results 3.2–3.5	EFDPR, ODI, sensitivity
21. Reporting biases	Discussion	Reviewer-cycle history
22. Certainty of evidence	Discussion	Pre-registered exact test failed to reject
23. Discussion	Discussion	Three caveats + three positives + outlook

Item	Reported in	Notes
24. Limitations	Discussion	Pilot scale, hand-curation, single guideline
25. Implications	Discussion	Outlook
26. Registration and protocol	Methods 2.6, prereg	`docs/prereg.md`
27. Support	Funding statement	No specific funding
28. Competing interests	Conflicts statement	None declared
29. Availability of data, code, materials	Statements	MIT licence, GitHub URL, Zenodo DOI at release
G1. Graph node space	Methods 2.3	(prior-state, biomarker) tuples
G2. Graph edge construction	Methods 2.3	Trial → edge from required-state node to post-trial-state node
G3. Guideline-tree encoding	Methods 2.4	16-node encoding in Supplementary Table S3
G4. Concordance algorithm	Methods 2.5	State-superset + biomarker tolerance + drug match + temporal precedence
G5. Tolerance levels	Methods 2.5	Strict / ESCAT / liberal, all pre-registered
G6. Reproducibility (seed + commit hash)	Methods 2.7	Seed 20260516; `v1.0.0` commit

Supplementary Table S2. Structured Trial Extractions

The hand-curated schema v1.0 mapping for each of the 16 pivotal trials is released as the machine-readable file `data/processed/trials_structured.json` in the repository. Each record contains the fields listed in Methods 2.2, the source primary publication, and the canonical ClinicalTrials.gov NCT identifier. The Round 1 internal review identified two NCT-misidentification errors in the original draft (NCT03997123 was incorrectly labelled as postMONARCH; NCT04032080 was incorrectly labelled as INAVO120); these were corrected to NCT05169567 (postMONARCH) and NCT04191499 (INAVO120), with TROPiCS-02 (NCT03901339) and OlympiAD (NCT02000622) added to the corpus.

Supplementary Table S3. ESMO HR+/HER2- mBC Decision-Tree Encoding

The hand-curated 16-node encoding of the ESMO 2024 update HR+/HER2- subset is released as `data/processed/esmo_decision_tree.json`. Each node records: node identifier, patient sub-state, biomarker profile, recommended drug classes with strength-of-recommendation grade, cited pivotal trials, and year of decision-tree introduction.

Supplementary Table S4. ODI Biomarker-Token Vocabulary

The operational-token vocabulary used to compute the operationalization discordance index is released as `analysis/06_compute_odi.py`. Each biomarker variable (HER2-low, ESR1mut, PIK3CAmut, AKT-pathway, prior-CDK4/6i) maps to a per-trial set of operational tokens drawn from the canonical primary publication.

Supplementary Text 1. Pre-registration deviations

The following pre-registration deviations are reported in the spirit of the pre-registration commitment in `docs/prereg.md`:

- **Corpus expansion from prototype (6 trials) to pilot (16 trials).** The pilot expansion was anticipated in the prereg and is not a deviation. Trial selection was based on the ESMO/ASCO mBC guideline reference lists; a systematic ClinicalTrials.gov search is deferred to the production extension.
- **Six NCT-identification corrections across Round 1 and Round 2 of internal review.** Round 1 corrected NCT03997123 (CAPItello-290 TNBC, misnamed postMONARCH) to NCT05169567 (real postMONARCH), and NCT04032080 (LY3023414+prexasertib TNBC, misnamed INAVO120) to NCT04191499 (real INAVO120). Round 2 corrected NCT02513394 (PALLAS adjuvant, misnamed PALOMA-2) to NCT01740427 (real PALOMA-2); NCT02675231 (monarcHER HR+/HER2+, misnamed MONARCH-3) was removed as out-of-scope and MONARCH-3 was instead encoded at NCT02246621 (whose previous label, MONALEESA-3, was itself wrong); MONALEESA-3 was added at NCT02422615 (real MONALEESA-3); and NCT02763566 (MONARCH plus postmenopausal, misnamed MONALEESA-7) was replaced with NCT02278120 (real MONALEESA-7). All NCT identifications in the present draft were verified against the ClinicalTrials.gov `briefTitle` and `conditionsModule` fields at the freeze date 2026-05-16. The full correction history is in the repository commit log.
- **TROPiCS-02 (NCT03901339) and OlympiAD (NCT02000622) added.** Two trials were added during the correction passes to populate decision nodes that the prereg's narrative implied but the prototype did not include: G14 (post-CDK4/6i+post-chemo sacituzumab) and G16 (gBRCAm PARPi). EMBRACA (talazoparib, NCT01945775), the parallel registrational trial for the gBRCAm node, is acknowledged in the guideline encoding but not included in this pilot corpus; it is queued for the production extension.
- **Scope deviation: G16 (gBRCAm) is line-agnostic and not strictly HR+/HER2-.** OlympiAD enrolled gBRCAm metastatic breast cancer regardless of HR or HER2 status, so G16 sits outside the prereg's strict HR+/HER2- inclusion. The primary analysis includes G16 with the line-agnostic state token `metastatic` handled as a wildcard; a 15-node sensitivity analysis (Results 3.5) confirms that the EFDPR finding is robust to G16 removal.
- **Secondary outcomes S2, S3 reported descriptively.** The prereg commits to a Benjamini-Hochberg correction at $q = 0.05$ across S1, S2, S3. In the present pilot S1 (ODI) is reported as a per-biomarker mean Jaccard distance with bootstrap CI; S2 (temporal lag) and S3 (subgroup coverage) are reported descriptively because the small corpus precludes a meaningful formal test.
- **Primary CI: Clopper-Pearson alongside bootstrap.** The prereg specified a 1,000-iteration bootstrap CI. We additionally report the Clopper-Pearson exact CI, which is the recommended primary CI in this small-n regime; both CIs are reported in Table 2.
- **Pre-registered exact-binomial test added as the formal primary test.** The prereg specified a one-sided exact-binomial test of $H_0 : \text{EFDPR} \leq 0.25$; an earlier internal draft incorrectly substituted a CI-overlap rule. The current draft reports the exact-binomial test as the primary test, and the CI-overlap rule has been removed.
- **Drug-class equivalence is an explicit table, not a prefix rule.** An earlier internal draft used a prefix-match rule for drug classes that was flagged as over-permissive in internal review; the current draft uses the explicit equivalence table in `analysis/05_compute_efdpr.py`.
- **NCCN sensitivity analysis deferred.** The prereg flagged NCCN as a sensitivity-analysis source; this was deferred to the production extension and is disclosed here.

- **Selection bias from guideline-list seeding.** The 16-trial pilot was seeded from the ESMO and ASCO guideline reference lists, which biases inclusion toward trials the guidelines already cite. The production extension will replace this seed with a systematic ClinicalTrials.gov query covering every phase 2/3 HR+/HER2- mBC trial registered between 2015 and 2026.
- **G6 state recoding between Round 3 and Round 4 of internal review.** The Round 1 and Round 2 drafts coded G6 (AKT-pathway recommendation) at `state="post-CDK46i"`; the Round 3 clinical-review pass flagged this as over-narrow relative to ESMO 2024's actual placement of capivasertib at the post-endocrine line with prior CDK4/6i permitted but not required, and the current draft codes G6 at `state="post-endo"`. Effect on numbers: strict EFDPR unchanged at 0.31 (G6 stays evidence-free under strict because CAPItello-291's AKT-pathway enters as a subgroup readout, not a biomarker inclusion criterion); liberal EFDPR fell from 0.25 to 0.19 because G6 is now liberal-supported by CAPItello-291's subgroup readout. The 15-node sensitivity tracked the change: liberal 0.27 $\rightarrow$ 0.20.